COSC 39: Theory of Computation

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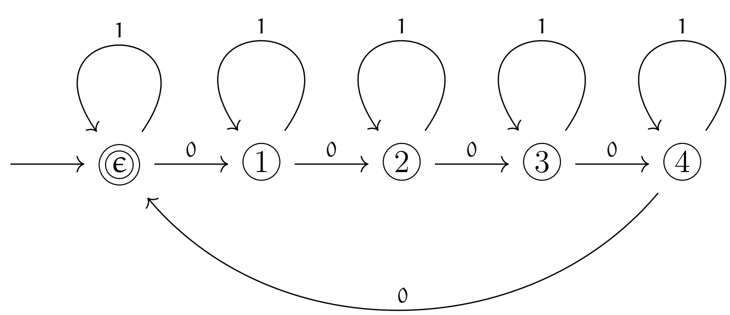
Credit Statement: Talked to Sophie Park'27.

Problem 1. Construct deterministic finite automata (DFAs) for each of the following languages. No formal proofs required, but explanation of your construction in a few English sentences would be helpful. Throughout the exercise we assume the alphabet to be $\Sigma = \{0, 1\}$.

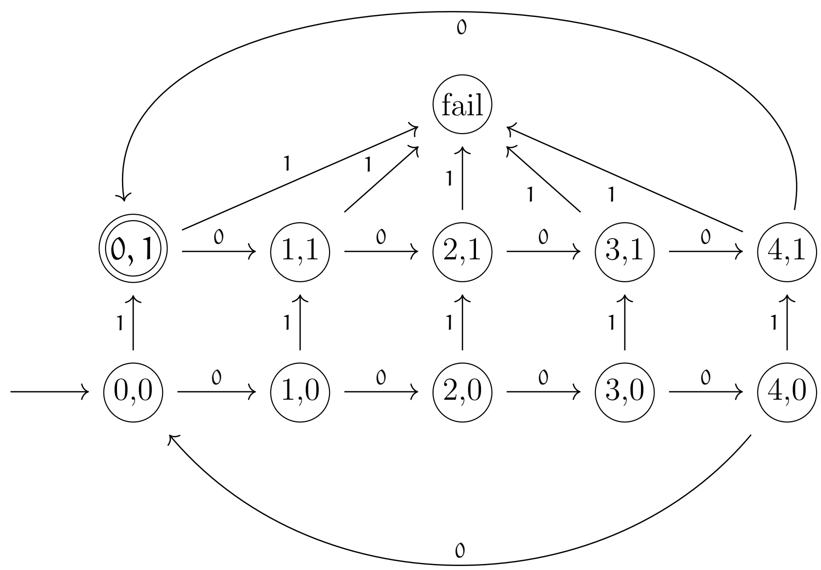
1. Set of all strings where number of 0s is divisible by 5
2. Set of all strings with exactly a single 1, and the number of 0s is divisible by 5.
3. Set of all strings containing 000 or 111 as subsequences.
4. Set of all strings not containing 000 nor 111 as subsequences.

Solution.

1. The DFA tracks the number of 0s seen so far modulo 5. Each state represents the current count of 0s mod 5; a 0-transition advances the state by 1 (mod 5), and a 1-transition leaves the state unchanged. The unique accepting state is 0 (the start state), since the empty string has zero 0s, and $0 \equiv 0 \pmod 5$.

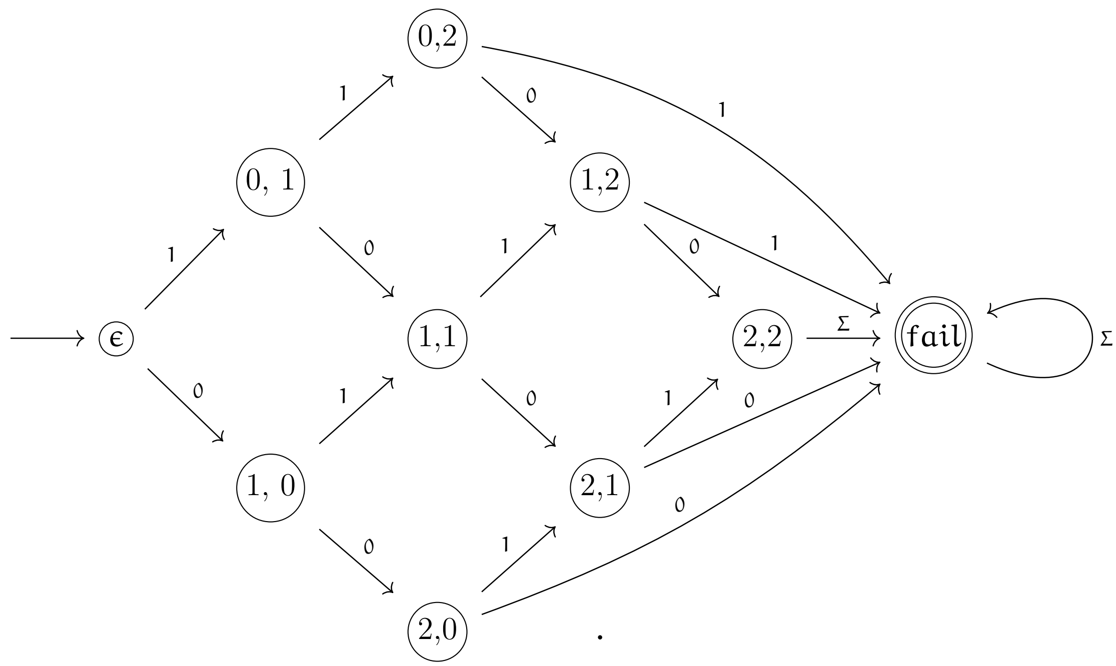


2. The DFA tracks two quantities simultaneously: the number of 0s seen mod 5, and whether zero or exactly one 1 has been seen. This gives a 5×2 grid of states (a, b) where a is the zero-count modulo 5 and b is the one-count. A 0-transition increments a by 1 (mod 5); a 1-transition increments b by 1. If $b \geq 2$ we get to a fail state. The unique accepting state is $(0, 1)$: zero 0s modulo 5 and exactly one 1.

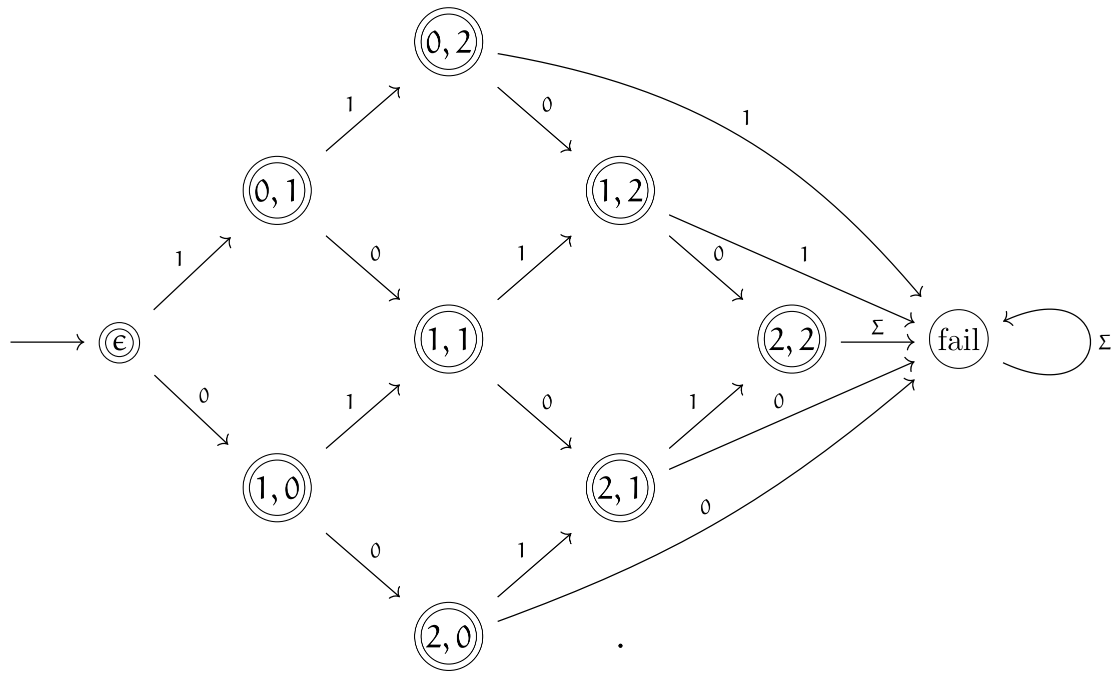


Note that the DFAs for parts (3) and (4) are complements of each other. Each state is a tuple (a, b) with $\mathsf{a}, \mathsf{b} \in \{\mathsf{0}, \mathsf{1}, \mathsf{2}\}$ respectively counting $\mathsf{0}$s and $\mathsf{1}$s seen so far. A $\mathsf{0}$-transition increments a (if $\mathsf{a} < 2$), and a $\mathsf{1}$-transition increments b (if $\mathsf{b} < 2$); once either counter would exceed 2, the machine enters a state representing "$\mathsf{000}$ or $\mathsf{111}$ has appeared as a subsequence." For part (3) this is the only accepting state; for part (4), which accepts strings containing neither subsequence, all states except this one are accepting.

3.



4.



Problem ⋆. To think about later:

5. Set of all strings that are binary representations of nonnegative integers divisible by 5.

6. Set of all strings whose reverses are nonnegative integers divisible by 5 in binary.